When Can a Bandit Trust Correlation?

Certified Pooling and Persistent Dynamics for Correlated Arms on the Web

Working long version, updated 12 October 2026 (revision 3)

Compiled from the `dependent_mab` repository, to be condensed into a WWW 2027 submission

Abstract

A short-video platform filling a daily trending slot faces arms that are correlated in two ways: videos watched by the same people tend to be liked by the same people, and a video's engagement today predicts its engagement for days. Platforms hold many graphs that seem to encode such structure, from co-engagement logs to social networks to hyperlinks, and graph bandits invite the learner to pool information along them. We ask when that trust is warranted.

We model reward as $f_{i,t} = \mu_i + z_{i,t}$, separating graph-certified permanent means from persistent, spatially correlated fluctuations. Certified pooling is valid under correctly specified innovation likelihoods and supplied valid widths; converting its realized information into regret bounds requires schedule-specific growth. Experiments support the mechanisms while exposing practical limitations. In a completed KuaiRec expansion with 120 held-out users and three subsets per user, oracle I-mf pooling has mean regret 0.640 times conservative matched KL-UCB, but 1.456 times Bernoulli Thompson sampling (TS); practical widths retain substantially less gain. Graph-free shrinkage explains much of spectral performance, although oracle pooling improves on a rewiring by 8.45%. Bursty persistence breaks iid confidence; innovation bounds can preserve coverage while remaining too wide to eliminate arms. A fresh AR(20) confirmation records 1/100 failures with known dynamics versus 40/100 with wrong-order, diagonalized dynamics. A chronological KuaiRec daily rerun reduces loss 49–54% against iid TS but does not establish an advantage over last observed value or over rewired/common-factor covariance. Full-year historical Wikipedia results depend on panel and budget. The Open Bandit Dataset supplies complementary mean-channel simulations in which graph-free empirical-Bayes TS is strongest, while an actual censored-log pilot is underpowered. The evidence supports conditional safety and information mechanisms, not universal algorithm superiority or a learned-model coverage theorem.

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Question	Theory that makes it testable	Answer from the evidence
Q1. Which correlation is real? (§3)	Mean restrictions and innovation covariance are separate; graph fit need not improve decisions	Collaborative graphs carry taste; sparse social coverage limits conclusions. Dynamic topology gains are unresolved after common-factor and rewiring controls.
Q2. How can a learner pool safely? (§4)	Supplied valid widths permit component rejection below the gap	Oracle pooling gains within a matched conservative family; practical widths are wide or incomplete. Strong graph-free shrinkage remains competitive.
Q3. What breaks when rewards persist? (§5)	Innovation confidence assumes correct dynamics; B2 quantifies iid failure	Bursty iid confidence can fail badly. Known-model coverage is compatible with its target; fitting and misspecification require separate audits. Valid elimination can make no decisions.
Q4. How should a learner act on persistent fluctuations? (§6)	Current reward, mean regret and PCS have different information objectives	Persistence helps selected panels, but simple last-value/EWMA rules capture much of the gain. Performance depends on panel, budget, model and comparator.

Table 1: Roadmap: four questions, the theory behind each, and the answers.

1 Introduction

A running example. Every day a short-video platform shows ten videos in a trending slot. Two facts shape the decision. First, videos are related: those watched by the same audience tend to be enjoyed by the same audience, so evidence about one video is evidence about its neighbours. Second, engagement persists: a video's completion rate today predicts tomorrow's, and shocks such as a holiday or an outage hit many videos at once. The platform also holds many graphs that might encode the first fact (co-engagement, matrix-factorization neighbours, creator links, tags, a social network) and must decide which to use, if any.

Two ways to get it wrong. A learner can *trust correlation that is not there*: pooling along a graph whose neighbourhoods do not share appeal biases estimates, and in the worst case leads to linear regret. Or it can *ignore that rewards persist*: confidence statements built for independent rewards are too narrow when consecutive observations of an item move together, so decisions that rely on them, such as dropping an item for good, go wrong. Both mistakes are easy to make on the web, where graphs are by-products rather than designed priors and where exposure is bursty: the same items fill a slate day after day.

Model. We write every reward as $f_{i,t} = \mu_i + z_{i,t}$. The *mean channel* restricts long-run appeal μ with a graph certificate; the *fluctuation channel* lets deviations z follow an autoregression with graph-correlated shocks. Graphs are a tool for encoding correlation in either channel, not the object of study.

Four questions. Table 1 lists the questions, the theory that makes each testable, and the answer the evidence gives. Each of Sections 3–6 follows the same pattern: what theory says, what it predicts, what we find, and what it means.

Contributions.

1. A two-channel model of correlated arms that contains iid bandits, static graph bandits and restless autoregressive bandits as special cases, and separates what a graph can encode (§2).
2. **Theory bridging the channels** (§5, proofs in Appendix A): conditional validity through exact innovation regression (B1); scalar independent-shock bounds (B1′) and iterated nuisance bounds for correlated shocks (B1″); a quantitative consecutive-sampling failure calculation for iid confidence (B2); and long-run information and restricted slate-contrast formulas (B3). Practical narrowness and learned-model validity are distinct open questions.
3. **Predictions tested against data.** Each theoretical statement yields a prediction that we test: certified gains track rejectable components (§4); uncertified pooling fails through dilution (§4); B2's variance inflation matches observed violation rates (§5); exploration pays with enough future observations per arm (§6).
4. **Evidence on three Web datasets** (KuaiRec, Wikipedia and OBD), plus controlled spatial/temporal simulations. Section 7 adds the completed 120-user expansion, chronological daily rerun, historical full-year extension, coverage stress tests and sparse-click studies, preserving the distinction between real panels, semi-synthetic feedback and actual logged feedback.

Provenance. Results proved in companion manuscripts in the repository are labelled [A] (graph alignment and certified pooling), [S] (spatiotemporal filtering and confidence), [P] (predictive and AR bandits) and [N] (a repository note on the selection index). Results first stated here are labelled [New].

2 Setting: two channels of correlation

There are N arms. In round t the learner pulls one arm, or a slate S_t of k arms, and observes

$$Y_t = \mu_{a_t} + z_{a_t,t} + \xi_t, \qquad z_t = \sum_{r=1}^{p} A_r z_{t-r} + \eta_t, \qquad \eta_t \sim N(0, Q_G), \qquad \xi_t \sim N(0, r). \tag{1}$$

Every arm evolves every round (restless); only pulled arms are observed.

Mean channel. Long-run means $\mu \in [0,1]^N$ are unknown. A graph supplies either a partition $\{C_c\}$ with certified diameters $\varepsilon_c \geq \max_{i \in C_c} \mu_i - \min_{i \in C_c} \mu_i$, or an energy certificate $\mu_c^\top L_c \mu_c \leq S_c^2$, which implies diameters through effective resistance [A].

Fluctuation channel. Deviations persist through A_r and are correlated across arms through Q_G, which can be shrunk toward a graph kernel. A shock common to all arms (a holiday) and a shock common to a community (a league's game day) are both correlation; only the second is something a graph can tell apart from a common factor.

Regret. *Mean regret* $\sum_t (\mu^\star - \mu_{a_t})$ measures learning of persistent appeal. *Dynamic regret* $\sum_t (\max_i f_{i,t} - f_{a_t,t})$ measures tracking of current rewards against an oracle that sees every arm's current value. With slates, both are summed over the slate against the oracle's top k.

Special cases. iid bandits: $A_r = 0$, Q_G diagonal, no certificate. Static graph bandits: $A_r = 0$ with a mean certificate. Restless AR bandits: no mean certificate. Spectral bandits regularize the mean channel without certifying it.

Graph	Side	Quotient (real / rewired)	Personal-taste corr. (real / random)	Choice cost, λ=10 (real / rewired)
I-mf (factorization)	item	0.756 / 0.961	**0.046** / 0.002	0.117 / 0.019
I-coeng (co-engagement)	item	1.098 / 1.302	**0.034** / 0.002	0.131 / 0.037
I-cat / I-tag (content)	item	0.908 / 0.985, 0.943 / 1.010	0.010 / 0.002	0.099 / 0.079, 0.100 / 0.054
I-coauthor (same creator)	item	0.805 / 1.006	0.004 / 0.002	0.040 / 0.184
U-coeng (co-engagement)	user	**0.649** / 0.791	**0.020** / 0.000	0.284 / 0.291
U-coauthor, U-coeng-idf	user	0.761 / 0.871, 0.748 / 0.860	0.018 / 0.000	0.296 / 0.294
U-mf (factorization)	user	0.850 / 0.907	0.016 / 0.001	0.300 / 0.298
U-cotag, U-cotime	user	0.842 / 0.884, 0.931 / 0.947	0.004, 0.003	0.299 / 0.297, 0.315 / 0.304
U-geo, U-feat, U-demo	user	0.946, 0.972, 0.988	0.007, 0.006, 0.002	0.350, 0.305, 0.328
U-soc (social; 47 edges)	user	not comparable	0.035 per edge	–

Table 2: KuaiRec: which graphs encode correlated appeal (reward = capped watch ratio, $k = 10$).

3 Q1. Which correlation is real?

3.1 What theory says

For the mean channel, a graph helps only if long-run appeal is smooth on it relative to the gaps that matter. Two facts sharpen this [A,N]. Smoothing on a complete graph shrinks all arms toward a common value and preserves every ranking; smoothing on an informative graph pulls an arm toward its neighbourhood's level, which can change the best arm (the misalignment example: $\mu = (1, 0.9, 0)$ with arms 1 and 3 linked makes arm 2 best after smoothing). So a graph can be *smoother than chance* and still *unsafe to smooth*. For the fluctuation channel, a graph helps only if shocks are correlated along its edges beyond what a single common factor explains.

Prediction 1. (a) Collaborative signals may align with personal taste more than profile or content features. (b) Fixed smoothing can change the best item even on a globally aligned graph. (c) A graph-specific decision gain should survive common-factor and rewiring controls; better shock likelihood alone need not improve ranking. These are empirical hypotheses rather than consequences of the confidence theorem.

3.2 Evidence: the mean channel on KuaiRec

KuaiRec contains a 1,411 user × 3,327 video matrix that is 99.6% observed. A larger sparse matrix (7,176 × 10,728) shares no (user, video) pair with it, so every graph is built leakage-free: user-side signals use only non-evaluation videos and video-side signals only non-evaluation users. We build sixteen graphs (Appendix B) and compare each with a degree-preserving rewiring. Table 2 and Figure 1 report three measures: a smoothness quotient (Laplacians scaled so a random graph scores about 1), the correlation of neighbours' personal taste after removing each user's activity and each video's popularity, and the cost to each user's best choice of smoothing true means with $(I + 10L)^{-1}$.

Findings.

- *Collaborative graphs carry taste* (Prediction 1a). Factorization and co-engagement neighbours agree on personal taste at 17–23 times the random-pair level; tag, category and creator graphs mostly group videos of similar popularity, and their advantage over the rewiring disappears once popularity is removed. Among user graphs, co-engagement is best; profile, location and demographic graphs are close to random. The social graph covers only 80 of 1,411 users. Rankings are stable across $k \in \{5, 10, 20\}$ and under a binary like-based reward (rank correlation 0.80).
- *Smoother than chance is not safe to smooth* (Prediction 1b). On item graphs, fixed smoothing costs more of a user's best-choice value on the real graph than on its rewiring (I-mf at $\lambda = 10$:

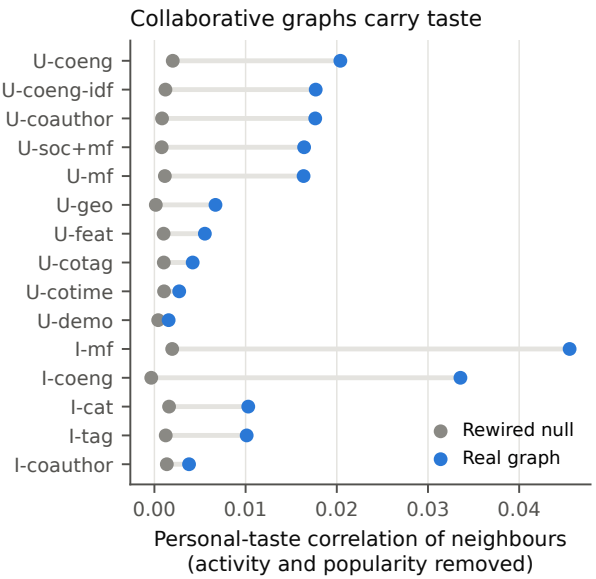


Figure 1: Personal-taste correlation of graph neighbours, real graph against its rewiring.

Panel	Graph	Shock corr.: edges / rewired / all	Within / across communities	Held-out log-lik per day: graph / rewired / common factor
KuaiRec daily	co-engagement	0.071 / 0.068 / 0.042	–	−323.0 / −323.3 / **−305.7**
Wikipedia, one domain	hyperlinks	0.256 / 0.250 / 0.218	–	52.88 / 52.85 / **53.76**
Wikipedia, six domains	hyperlinks	**0.273** / 0.051 / 0.051	0.320 / 0.000	**22.16** / 21.67 / 21.92

Table 3: Do graphs carry correlated shocks? Held-out log-likelihood scores AR residuals of a 120-day (Wikipedia) or remaining-days (KuaiRec) window under each innovation-covariance target fitted on the preceding window, averaged over three origins. Domain blocks score 22.13 on the six-domain panel.

0.117 against 0.019). The rewiring smooths each video toward a random sample of others, which acts like uniform shrinkage; the real graph pulls videos toward their community, which reorders the top. The exception (I-coauthor) makes small cliques of near-identical videos.

- User graphs are dominated by a shared popularity component: users' rows correlate at 0.38 even for random pairs.

3.3 Legacy diagnostics: the fluctuation channel on three panels

These original KuaiRec and 150-article Wikipedia panels used future interaction inputs or retrospective cohort/link selection. Their covariance diagnostics are retained as exploratory mechanism evidence. The chronological KuaiRec and historical 24-article Wikipedia updates in Section 7 govern current policy conclusions.

Table 3 compares shock correlation along graph edges, along rewired edges and across all pairs, and the held-out likelihood of the innovation covariance under each graph target, on three daily panels: KuaiRec (253 videos, 63 days), Wikipedia programming-language articles (150 articles, 2022–2025) and Wikipedia articles from six communities (NFL, NBA, MLB, Premier League, countries, programming languages; 25 each). The six-community panel was designed after the first two returned a null, to test Prediction 1c directly.

Findings.

- On KuaiRec and on single-domain Wikipedia, shocks are correlated, but through a platform- or domain-wide factor: graph edges are no more correlated than rewired edges, and a common-factor covariance fits held-out shocks best.
- With six communities, the picture reverses (Prediction 1c). Shocks correlate at 0.32 within a community and 0.00 across; 84% of hyperlinks stay within a community, so hyperlinked pairs correlate at 0.273 against 0.051 for rewired pairs. Hyperlinks give the best held-out fit at all three origins, matching an oracle community-block graph (22.13).

Answer. Collaborative graphs carry useful preference signal in these data, but alignment does not certify safe smoothing. Social-graph coverage is too sparse for a broad conclusion. For fluctuations, compare topology against a common factor and rewiring: stronger shock fit does not by itself establish better decisions.

4 Q2. How can a learner pool safely?

4.1 What theory says

Certified pooling [A]. SP-UCB keeps an arm upper bound $u_i = \bar{Y}_i + b(n_i)$, $b(n) = \sigma\sqrt{2\ell/n}$, and a pooled component bound, selects the component maximizing $U_c = \min\{\max_{i\in C_c} u_i,\ \hat{p}_c + b(N_c) + \varepsilon_c\}$ and pulls its best arm.

Theorem 1 (Alignment-sensitive regret [A])**.** *With valid certificates, with probability at least* $1-\delta$, $\mathcal{R}_T \le \sum_{c\in\mathcal{O}} A_c + \sum_{c\notin\mathcal{O}} \min\{A_c, H_c\}$, *where* $A_c = \sum_{i\in C_c:\Delta_i>0}(\Delta_i + 8\sigma^2\ell/\Delta_i)$ *is the cost of exploring the component arm by arm and* $H_c = (\Gamma_c + \varepsilon_c)(1 + 8\sigma^2\ell/(\Gamma_c - \varepsilon_c)^2)$ *is the cost of rejecting it as a whole, finite only when its width* ε_c *is below its gap* Γ_c.

Graph geometry, not diameter, sets the value of a certificate: a path and a clique with identical diameters and energy budgets differ in information cost by a factor growing with the component size, and the cost of loosening a certificate grows with the resistance radius [A]. A certified Laplacian comparison $(1-\eta)L \preceq \hat{L} \preceq (1+\eta)L$ allows estimated graphs by inflating the energy radius to $\sqrt{1+\eta}\,S$ [A].

Prediction 2. (a) Certified pooling gains only through components whose width is below their gap: gains should rise with the share of such *rejectable* components. (b) Pooling with zero width (no certificate) fails when the optimal arm's component has a lower average than another component, since the pooled bound then rejects the optimal component; larger optimal components dilute more. (c) Average gains come mostly from shrinkage, so rewired graphs should do about as well on average.

4.2 Evidence

Legacy design (Setting A). For each of 60 test users, 300 videos are sampled and each item graph is rebuilt on them; SP-UCB uses a 20-component spectral partition; rewards are Bernoulli with the user's data-derived means; $T = 20{,}000$; common random numbers across policies. The original Setting A v2 used 30 users × 2 subsets. It is now superseded for headline performance by the completed 120-user × 3-subset expansion in Section 7; the older mechanism tables below retain their original cohort.

Share of suboptimal components rejectable (oracle certificate)	Instances	Median SP-UCB / UCB1	Interquartile range
$\leq 40\%$	54	0.84	0.78–0.91
40–60%	67	0.73	0.63–0.79
60–80%	119	0.52	0.41–0.61
$> 80\%$	120	0.40	0.30–0.83
Descriptive Spearman correlation: -0.48; users recur across graph instances.			

Table 4: Certified gains track the theorem's rejectability condition (Prediction 2a).

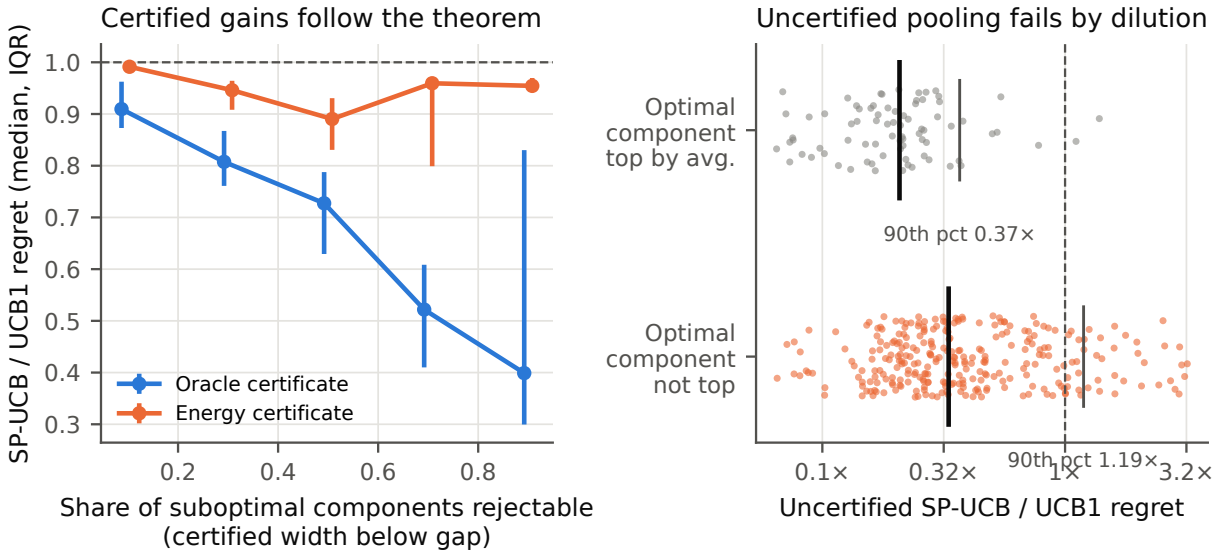


Figure 2: Left: certified gain against the share of rejectable components (median and interquartile range). Right: uncertified pooling, split by whether the optimal arm's component is the top component by average; failures (ratio above 1) occur almost only when it is not.

Findings.

- *Gains follow the theorem* (Prediction 2a; Table 4, Figure 2). The median regret ratio of oracle-certified SP-UCB to UCB1 falls from 0.84 when at most 40% of suboptimal components are rejectable to 0.40 when more than 80% are. Energy certificates are wider (energy-to-gap ratio about 5 on every graph) and rarely make components rejectable: in 268 of 360 instances fewer than 40% are, and their median gain is only 2–11% in every bin. The widest bin for oracle certificates ($> 80\%$) has a long upper tail (75th percentile 0.83), driven by graphs with many tiny components.
- *Uncertified pooling can fail by dilution* (Prediction 2b). In this legacy cohort, the top-component-by-average group has median 0.21× UCB1 and 90th percentile 0.37×; otherwise the 90th percentile is 1.19×. This association is not a safety certificate. Failures concentrate on some informative creator/category graphs, but zero failures in a particular rewiring sample do not imply safety on random graphs. The completed expansion and OBD examples in Section 7 retain controls and objective-specific failure diagnostics.
- *Average gains are shrinkage* (Prediction 2c). SpectralUCB at $\lambda = 10$ regrets 0.697× UCB1 on I-mf and 0.695× on its rewiring; spectral Thompson sampling 0.611× against 0.610×. Uncertified pooling with KL bounds is best on average (0.67× KL-UCB) and does as well on the rewired graph. In Setting B (users arrive at random, 300 users × 100 videos), shrinking toward the

SP-KLUCB certificate (I-mf)	Regret / KL-UCB (mean / 90th pct.)	Components valid	Mean width
Oracle (true within-component range)	0.78 / 1.05	100%	0.36
Per-user calibrated, 75th percentile	0.88 / 1.05	73%	0.40
Population calibrated, 50th percentile	0.81 / 1.05	52%	0.30
Per-user calibrated, 90th percentile	0.96 / 1.05	90%	0.60
Population calibrated, 90th percentile	1.00 / 1.05	93%	0.71
Energy (from the graph)	0.98 / 1.05	100%	0.80
None (zero width)	0.67 / 1.32	0%	0

Table 5: Legacy certificate gap: 30 users × 2 subsets, KL bounds. Entries average paired instance ratios; KL-UCB uses the matched fixed confidence level. Per-user diagnostics use the true user level while policies use a paid noisy warm-up. Table 11 reports the completed expansion and distinguishes ratios of mean losses.

population mean cuts regret 27% and no user graph beats it; user pooling without a certificate is 39% worse than none, with an oracle certificate 6% better.

The certificate gap. Table 5 shows the original practical bottleneck. Truth-supplied energy widths are valid but wide. Population and level-matched quantile widths have empirical component coverage below one; a calibration quantile is not a simultaneous guarantee for a new user. The completed expansion retains this tradeoff: level-matching improves performance within the conservative KL-UCB family but neither matches oracle widths nor establishes an advantage over Bernoulli TS.

Answer. A valid width below a component gap makes certified group rejection possible. Incorrect widths can cause dilution failures, while wide widths remove much of the gain. Oracle pooling helps the matched conservative baseline; practical calibration remains empirically incomplete, and graph-free shrinkage is a strong comparator.

5 Q3. What breaks when rewards persist?

5.1 What theory says

Certified pooling relies on confidence bounds. With persistent, restless rewards, consecutive observations of an arm are correlated, and an arm's current value is partly predictable from the past. The *innovation regression* of [S] removes the predictable part exactly.

Theorem 2 (Predictable innovation experiments [S])**.** *Conditional on a candidate mean μ, a Kalman recursion that does not depend on μ gives, for the chosen arm, $R_t = u_t^\top \mu + \epsilon_t$ with predictable u_t and $\epsilon_t \mid \mathcal{H}_t \sim N(0,1)$. With $J_T = H + \sum_t u_t u_t^\top$, $\hat{\mu}_T = J_T^{-1} \sum_t u_t R_t$ and a self-normalized width β_T, $|v^\top(\hat{\mu}_T - \mu)| \le \beta_T \|v\|_{J_T^{-1}}$ for all T and all v with probability at least $1-\delta$.*

Building SP-UCB's bounds from this regression gives four results (statements here, proofs in Appendix A).

Theorem 3 (B1: validity and regret [New])**.** *With valid certificates and correctly specified dynamics, the innovation-based arm and component bounds hold at every round with probability at least $1-\delta$, under any adaptive restless schedule. If each pull adds information at least ι to the relevant contrast, Theorem 1 holds with $8\sigma^2\ell$ replaced by $4\beta_T^2/\iota$; for independent AR(1) arms with $0 \le \varphi < 1$, $\iota \ge (1-\varphi)^2/(V+r)$ per scalar reading. Correlated partial schedules need a separately justified growth condition.*

B1 is valid but wide: its width carries an N-dimensional log-determinant (about 12 standard errors with 20 arms and $T = 20{,}000$, against about 5.5 for a per-arm bound).

Proposition 4 (B1′: independent shocks [New])**.** *If Q_G is diagonal, the innovation information is diagonal and every arm and component statistic is a scalar martingale; one-dimensional mixture bounds with a union over the $N + M$ statistics are valid under the stated model. Their practical width depends on the realized information and does not guarantee useful elimination at a given horizon.*

Proposition 5 (B1″: correlated shocks [New])**.** *With correlated shocks, other arms' means enter each arm's regression as bounded nuisances $\sum_{j\neq i} G_{ij}(\mu_j - m_j)$, $G = J - \alpha I$. On the event that all scalar mixture bounds hold, plugging in intervals that contain the truth yields intervals that contain the truth, so iterating from $[0, 1]$ is valid. Writing $\mu_i = \theta_c + d_i$ with $|d_i| \leq \varepsilon_c/2$ gives a valid pooled component bound; with independent shocks it reduces to SP-UCB's.*

Proposition 6 (B2: iid certificates under persistence [New])**.** *For b consecutive observations of a stationary AR(1) arm with persistence $0 \leq \varphi < 1$, marginal variance V and no sensor noise, $\mathrm{Var}(\bar{Y}_b) = (V/b)f(b, \varphi)$, where $f(b, \varphi) = \frac{1+\varphi}{1-\varphi} - \frac{2\varphi(1-\varphi^b)}{b(1-\varphi)^2}$. The radius $\sqrt{2V\ell/b}$ therefore has exact per-check miscoverage $2\Phi(-\sqrt{2\ell}/\sqrt{f(b, \varphi)})$. Multiple separated blocks require their cross-block autocovariances; the same expression is only an approximation when those terms are negligible.*

Proposition 7 (B3: long-run substitution [New])**.** *For a continuously observed latent field with common AR coefficients, the long-run covariance is $\Omega = Q_G/c^2$, $c = 1 - \sum_r a_r$, and Gaussian mean information has the corresponding inverse limit. Independent sensor noise adds rI to the observed long-run covariance. With zero sensor noise, a continuously observed subset S uses Q_{SS}/c^2 and precision $c^2(Q_{SS})^{-1}$, not $c^2(Q_G^{-1})_{SS}$. Two arms with equal marginal innovation variance q and matched coefficients have latent long-run contrast variance $2q(1 - \rho)/c^2$. With unequal variances or coefficients, use $\Omega_{ii} + \Omega_{jj} - 2\Omega_{ij}$ and add observation noise; arbitrary partial restless schedules need their conditional likelihood. This information substitution is not an unconditional transfer of all count-based regret constants.*

Prediction 3. (a) Bursty persistence can invalidate iid confidence; interleaving may reduce the effect but is not a universal coverage remedy. (b) Correct-model innovation bounds obey their nominal probability statement; zero simulated violations are not a theorem of zero failure. (c) Narrower constructions may improve on the joint ellipsoid, but useful rejection and regret must be measured separately. (d) Invalid confidence can lead to irreversible errors when the optimal arm or component is under-covered.

5.2 Evidence

Design. Twenty arms in four components of five, component bands of width $\varepsilon_c = 0.1$, best mean 0.55; AR(1) deviations with $\mathrm{Var}(z) = 1$ and $\varphi \in \{0, 0.5, 0.9, 0.97\}$; independent shocks or shocks correlated along a path; observation sd 0.1; $T = 5{,}000$; 20 seeds with shuffled arm order. Exposure is one arm at a time or blocks of 25 consecutive rounds (a daily slate or a batched update).

Findings.

- *B2 predicts the failure* (Prediction 3a; Table 6, Figure 3). Blocks of 25 inflate the sample-mean standard deviation by 1.7, 3.5 and 4.4 at $\varphi = 0.5$, 0.9, 0.97; the iid certificate is then violated in 0%, 85% and 90–95% of runs. One-at-a-time exposure interleaves arms (repeat pulls about 20 rounds apart, correlation $\varphi^{20} \approx 0.54$ at 0.97), so the inflation is small and the conservative union-bound level still covers.

φ	B2 sd inflation (blocks of 25)	B2 per-check miscoverage	iid: runs violated	iid: rounds violated	B1″: runs violated
0	1.00	$< 10^{-6}$	0%	0%	0%
0.5	1.68	0.1%	0%	0%	0%
0.9	3.49	11%	85%	58–70%	0%
0.97	4.42	21%	90–95%	83–85%	0%

Table 6: B2's prediction against observed violations (blocks of 25; both shock structures). With one-at-a-time exposure, iid certificates were never violated at any φ. Run-level rates exceed per-check rates because each run checks 24 statistics every round.

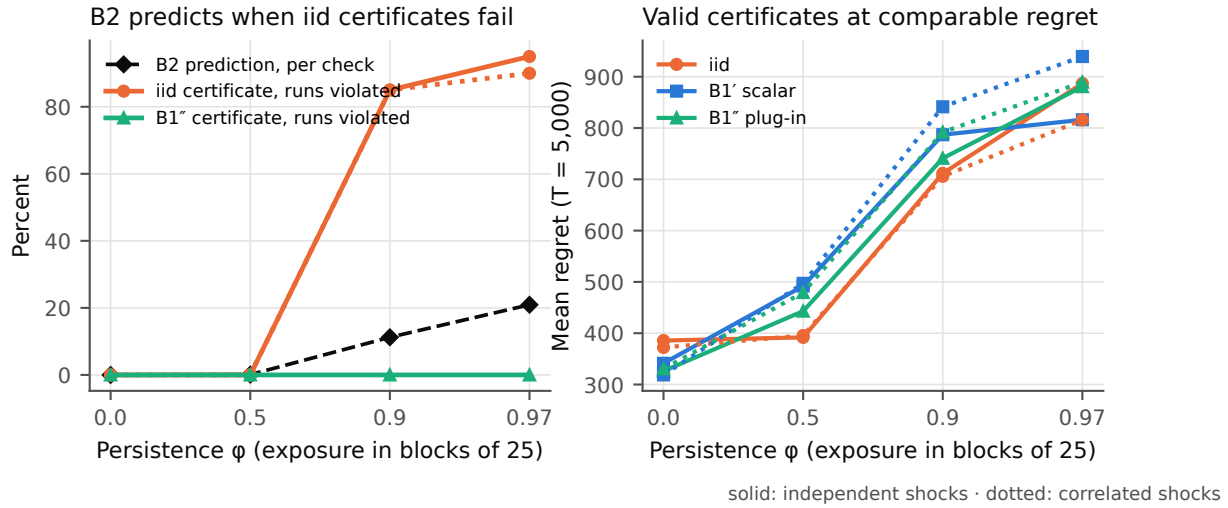


Figure 3: Left: B2's predicted per-check miscoverage and the observed share of runs with a violated iid certificate, blocks of 25. Right: mean regret of SP-UCB with iid, B1′ and B1″ certificates.

- *Correct-model confidence is supported conditionally* (Prediction 3b): the original sweeps record no innovation violations under their tested model assumptions. The fresh confirmation records 1/100 known-model failures and 5/100 fitted-model failures; wrong order plus diagonal innovations gives 40/100. Section 7 separates these from the earlier sweeps. B1′ is guaranteed only for independent shocks.
- *Narrower constructions help selected toy settings* (Prediction 3c; Table 7), but their advantage does not generalize to real-calibrated dynamics. At day 730 in the latest Wikipedia-derived study, median B1″ width is 1.74 against a best–second gap of 0.135, and innovation elimination retains every arm. Validity and useful power are different requirements.
- *Irrevocable decisions inherit the failure* (Prediction 3d). Successive elimination with iid certificates wrongly eliminated the best arm in 35–90% of bursty high-persistence runs, against a nominal 5%; with innovation certificates, never. The price is caution: at these gaps (0.1–0.4 against unit fluctuation variance) valid certificates eliminate almost nothing within 5,000–20,000 rounds.

Answer. Confidence must account for the actual temporal sampling schedule. Bursty iid bounds can fail and can trigger irreversible errors. Innovation bounds have conditional guarantees under correct dynamics and valid supplied widths; estimating the model or misspecifying it needs a separate audit. Even empirically well-covered bounds can be too wide to eliminate arms or lower finite-horizon regret.

Exposure	φ	Independent shocks: iid / B1 / B1′	Correlated shocks: iid / B1′ / B1″
One at a time	0	353 / 753 / **295**	357 / 317 / **313**
One at a time	0.5	457 / 752 / **396**	460 / 383 / 412
One at a time	0.9	533 / 759 / 564	642 / 646 / **606**
One at a time	0.97	649 / 755 / 679	693 / 683 / 760
Blocks of 25	0	380 / 766 / **338**	372 / 319 / 330
Blocks of 25	0.5	406 / 916 / 510	395 / 497 / 479
Blocks of 25	0.9	629† / 1,023 / 818	706† / 841 / 792
Blocks of 25	0.97	853† / 1,034 / 828	816† / 939 / 889

Table 7: Mean regret ($T = 5{,}000$). † iid certificate invalid in 75–95% of runs. B1′ has no guarantee under correlated shocks; B1″ does. The independent-shock columns use unshuffled worlds and are from a separate sweep.

6 Q4. How should a learner act on persistent fluctuations?

6.1 What theory says

Joint filter. Stacking long-run means and AR lags gives one Gaussian state that every observation updates for every arm and lag [S]. We verify the implementation against dense conditioning on the full node-time covariance.

Proposition 8 (Innovation lower bound [S,P])**.** *With stationary covariance K_0 and innovation covariance Q_G, every causal policy has dynamic regret at least $T\{\mathcal{G}(\mu, K_0) - \mathcal{G}(\mu, K_0 - Q_G)\}$, $\mathcal{G}(\mu, K) = \mathbb{E}\max_i(\mu_i + Z_i)$.*

Which uncertainty to explore. Full-state randomization can pay a cost for uncertainty with little useful continuation value. One-step predictive sampling [P] uses $B = CV^{-1}C^\top$, with $C = \mathrm{Cov}(f_t, f_{t+1})$ and $V = \mathrm{Var}(f_{t+1})$, to emphasize uncertainty related to the next field. This differs from subtracting fresh innovation covariance from a forecast, and from the older KuaiRec rule that perturbs only permanent means. Current innovations can influence later states through the AR dynamics; their information value is not universally zero. The full impulse profile and remaining horizon matter, and these heuristics have no general optimality guarantee.

Prediction 4. (a) Temporal inference can improve on iid decisions, but should also be compared with last value, EWMA, discounting and static ranking. (b) Full-state and predictive sampling may trade off differently by objective and persistence. (c) The value of exploration depends on future information, gaps and initialization. (d) Any topology claim must survive a matched temporal rule, a common factor and rewired graphs. These are hypotheses tested in specific regimes, not uniform performance theorems.

6.2 Evidence: a controlled benchmark

Design. 100 arms on a 10×10 grid; AR(20) coefficients proportional to $e^{-r/6}$ summing to 0.97; shocks correlated along the grid ($(I + 5L)^{-1}$, unit diagonal); smooth long-run means; observation sd 0.3; $T = 2{,}000$; 8 seeds; variants with weak persistence (0.6) and with independent shocks. Model-based policies know the dynamics; means are unknown.

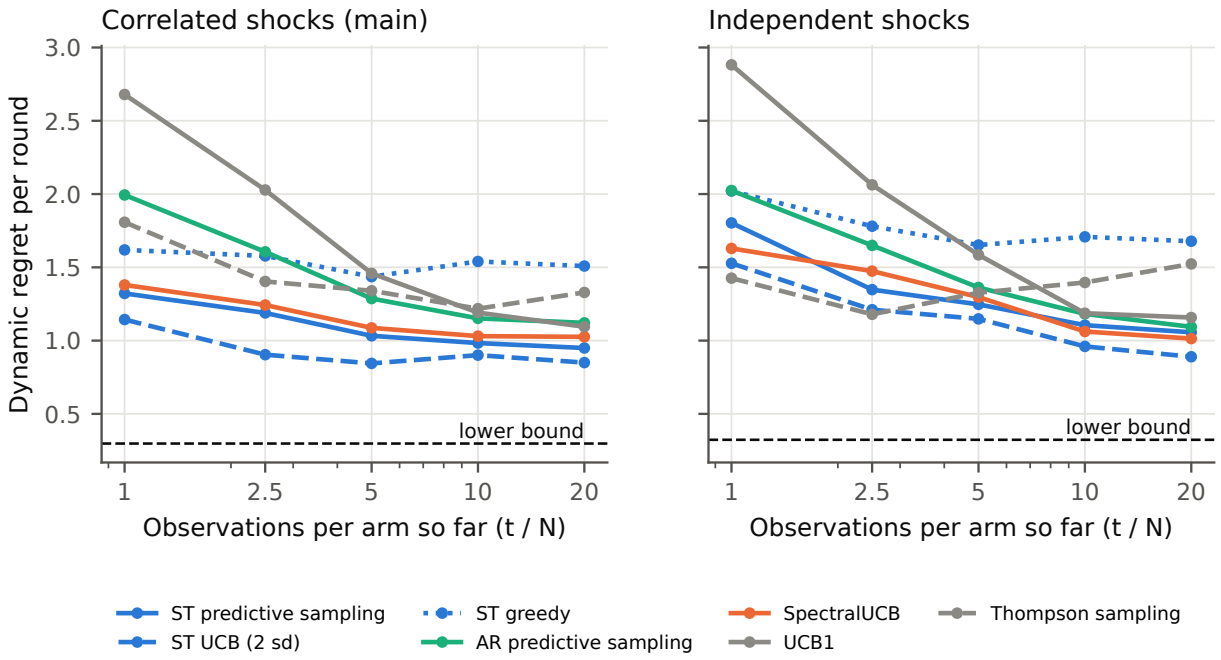


Figure 4: Benchmark: average dynamic regret over the first t rounds (main configuration, left; independent shocks, right). Dashed: innovation lower bound.

Findings. Joint predictive sampling, which has no tuning parameter, cuts dynamic regret 27% against Thompson sampling and 13% against UCB1, closing 37% of the gap to the lower bound (tuned UCB closes 46%; its multiplier was chosen on the same runs). Mis-targeted exploration is as harmful as having no model: greedy, current-state Thompson sampling and the theory-constant UCB all lose to UCB1 (Prediction 4b). Predictive sampling leads greedy at every horizon in Figure 4, because priors are weak and each arm receives 20 observations. Correlated shocks add 15% on top of persistence (joint against persistence-only predictive sampling); with independent shocks the gain disappears (Prediction 4d). Here fluctuations dominate the decision: their standard deviation (1.0) is twice the spread of long-run means (0.5). With weak persistence the lower bound is already high and all gains shrink.

6.3 Legacy exploratory evidence: two web platforms

The following panels are retained to explain the original hypotheses. The daily KuaiRec graph used all interaction dates and the cohort conditioned on presence throughout the observation period. The 150-article Wikipedia cohorts and link snapshots were selected retrospectively. Their apparent policy gains are not the current chronological headline; Section 7 supplies the corrected KuaiRec and historical Wikipedia comparisons.

Designs. *KuaiRec daily:* 253 videos observed on all 63 days; reward the logit daily complete-play rate; ten slots per day; models fitted on the first $F \in \{28, 35, 42\}$ days; a sparse-history variant sees only ten videos per fit day. *Wikipedia:* log daily views of 150 articles; ten featured slots per day; models fitted on a rolling 365-day window and tested on the next 120 days at three origins (January 2023, January 2024, December 2024); one-domain and six-domain panels. All panels assume that

Policy	Models	Main	Independent shocks	Weak persistence
ST UCB, 2 sd (tuned)	both channels	**0.850**	**0.890**	1.856
ST predictive sampling	both channels	0.950	1.057	1.568
SpectralUCB	mean channel only	1.026	1.014	1.953
UCB1	nothing	1.095	1.157	1.867
AR predictive sampling	persistence only	1.121	1.094	**1.525**
ST current-state TS	both channels	1.258	1.452	2.246
Thompson sampling	nothing	1.329	1.523	1.675
ST greedy	both channels	1.509	1.678	1.932
Innovation lower bound		0.298	0.323	1.276

Table 8: Benchmark: dynamic regret per round at $T = 2{,}000$.

Policy (regret per day)	KuaiRec full	KuaiRec sparse	Wikipedia one domain	Wikipedia six domains
Model, greedy	2.41	3.93	0.441	1.327
Model + drift, greedy	**2.36**	–	**0.376**	1.257
Model, UCB 1 sd	2.57	**3.59**	0.436	**1.205**
Model, predictive sampling	4.04	6.85	0.407	1.470
Temporal-only model, greedy	2.44	–	0.448	1.309
Last observed value	2.88	–	0.486	1.434
iid Thompson sampling	3.90	6.35	0.553	1.588
Graph in shock covariance: real / rewired / none	2.63 / 2.64 / 2.62*	–	0.441 / 0.441 / 0.441	1.327 / 1.307 / 1.299

Table 9: Web platforms, regret per day. KuaiRec: mean over $F \in \{28, 35, 42\}$ (5 seeds for stochastic policies). Wikipedia: mean over three origins and 5 noise seeds. Graph row: greedy policies, except *persistent sampling on KuaiRec. On six domains, community blocks give 1.290 (greedy) and 1.456 (predictive sampling, against 1.500 with the common factor).

showing an item does not change its organic engagement.

KuaiRec daily. Deviations persist (lag-1 autocorrelation 0.78) and are as large as the spread of long-run levels (sd 0.46 against 0.54). The panel is one upload cohort (median age 2 days at the start), whose engagement falls during days 21–42 (mean slope -0.023 per day) with heterogeneous level shifts of about 0.5, and then flattens. A stationary-mean model fitted through day 42 therefore reverts toward stale early-life levels, which is why the last observed value is the best rule at $F = 42$ (0.93 regret per day against 1.31). A small likelihood-selected drift in the level gives the best average (2.36).

Wikipedia. On one domain, attention persists with a weekly cycle (lag-1 autocorrelation 0.72, lag-7 0.74) and the oracle's top ten is stable (1–6% daily turnover, gap between the 10th and 11th article 0.07–0.21). Most regret comes from one window: in January 2024, 12.6% of each day's top ten were articles outside the fit window's top ten, against 1.3% a year earlier. The drifting-level filter cuts regret 32% against iid Thompson sampling and predictive sampling 26%. On six domains, rankings turn over far faster (22% per day; gap 0.03) because game days rotate which teams lead. Every model beats iid Thompson sampling (best: -24%), and mild optimism (UCB, 1 sd) is the best policy.

Does the graph help decisions where it carries shocks? Only slightly. On six domains, where hyperlinks carry community shocks and fit held-out shocks best (Table 3), putting them into the shock covariance changes regret by -2.0% under predictive sampling ($\pm 1.3\%$ s.e., 15 paired

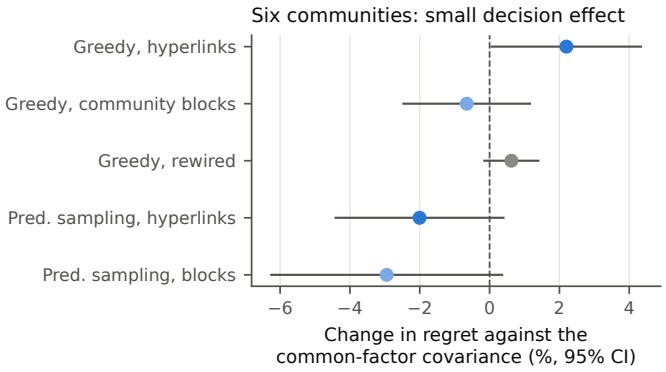


Figure 5: Six-domain Wikipedia: change in regret from a graph-shrunk shock covariance against a common-factor covariance, paired by origin and noise seed (95% intervals).

Environment	Future obs. per arm	10th–11th gap	Daily top-10 turnover	Fluctuation / level sd	Pred. sampling / greedy	UCB 1 sd / greedy	Best model / iid TS
Benchmark	20	–	–	2.0	**0.63**	0.73	0.71
Wikipedia, one domain	8	0.15	3%	0.39	**0.92**	0.99	0.74
Wikipedia, six domains	8	0.03	22%	0.52	1.11	**0.91**	0.76
KuaiRec daily, full history	1.1	0.02	33%	0.85	1.67	1.07	0.62
KuaiRec daily, sparse history	1.1	0.02	33%	0.85	1.74	**0.91**	0.56

Table 10: When exploration pays. Ratios below 1 favour the exploring policy over greedy use of the same model.

runs) and by +2.2% under greedy (±1.1%); community blocks give −2.9% and −0.7% (Figure 5). Neighbours predict shocks about as well as in the benchmark (R^2 0.36 against 0.46), so the difference lies elsewhere: on Wikipedia, which articles lead is set mostly by long-run levels (fluctuation sd 0.48 against level spread 0.91), whereas in the benchmark fluctuations dominate (sd 1.0 against 0.5). Sharper beliefs about today's shocks rarely change the top ten when levels decide it.

6.4 Synthesis: when does exploration pay, and when does the graph help?

Table 10 lines up the five environments.

What the synthesis shows.

- *Temporal information helps in selected environments*: the legacy panels suggested 24–44% lower loss than iid TS. The chronological KuaiRec rerun supports persistence-aware rules but does not distinguish the full model from last value; full-year Wikipedia gains depend on panel and budget. The original universal headline is therefore withdrawn.
- *How much to explore depends on observations per arm and on gaps* (Prediction 4c, refined). Heavy exploration of persistent uncertainty (predictive sampling) beats greedy only where arms receive many observations and the top is well separated or dominated by fluctuations (benchmark, one-domain Wikipedia). With near-ties and fast turnover it shuffles near-equal items at a cost, even with eight observations per arm (six-domain Wikipedia). Mild optimism helps wherever priors are weak or rankings move (benchmark, six-domain Wikipedia, sparse KuaiRec) and is roughly neutral otherwise; with strong priors and about one observation per arm (full-history KuaiRec), greedy is best.
- *The graph helps the fluctuation channel in proportion to how much fluctuations decide the ranking* (Prediction 4d, refined): 15% in the benchmark, where fluctuations dominate; at most ±3% on six-domain Wikipedia, where links carry shocks but levels decide the top ten; nothing where

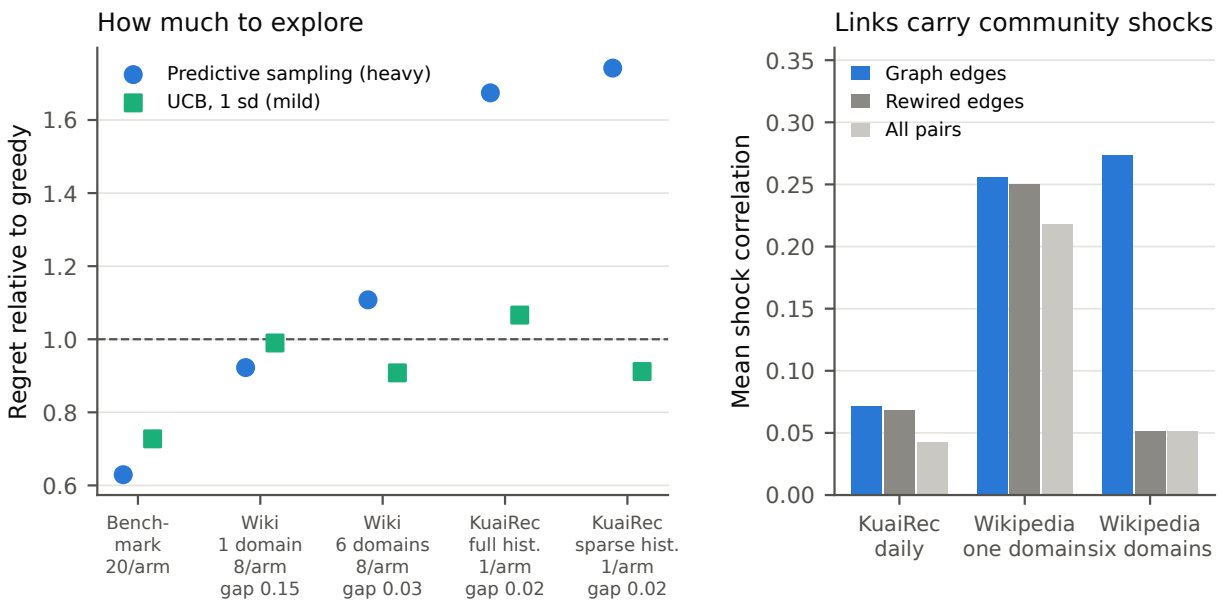


Figure 6: Left: predictive sampling relative to greedy use of the same model, against the number of future observations each arm receives. Right: value of the graph in the shock covariance (held-out log-likelihood gain over a common-factor model) on the three daily panels.

shocks are not community-specific.

Answer. Use temporal information when it improves the intended objective, and retain simple persistence and static controls. The chronological KuaiRec filter improves on iid TS but does not establish a gain over last value. Historical Wikipedia results vary by panel and budget. A better covariance fit or a joint-versus-iid gain alone does not establish that the actual graph topology improves decisions.

7 Updated empirical evidence: completed 10–11 October runs

This section supersedes earlier empirical headlines where protocols or comparators have changed. It incorporates the completed 120-user KuaiRec expansion, the chronological daily rerun, the full-year historical Wikipedia extension, general-AR coverage confirmation, real-calibrated certificates, and the Open Bandit Dataset (OBD). The experiments finished on 10–11 October; this manuscript update is dated 12 October. Legacy numerical studies and derivations remain useful under their original assumptions.

What the evidence tests. Static KuaiRec and the main OBD pooling study simulate Bernoulli feedback with real-data-derived means. The daily panels emulate a monitoring budget on an observed field. Controlled worlds identify permanent means, state-oracle regret and coverage separately. The OBD censored-log pilot uses actual randomized logged feedback. These are distinct estimands; no aggregate regret or probability-of-correct-selection (PCS) is formed across them. Known-model, oracle-width and fitted-model results also remain distinct. All comparisons below are exploratory and intervals are pointwise, without adjustment for the many policies and configurations examined.

7.1 Mean-channel pooling: the completed 120-user expansion

Setting A v2 now comprises 120 held-out users and three independently seeded video subsets per user: 360 distinct user–subset instances, 300 videos per instance and 20,000 pulls. The main phase contains 10,440 policy trajectories; the subsequent per-user-calibration phase adds 1,440 trajectories on the same instances. Thus the 720 logged phase completions are not 720 independent data instances. Feedback is simulated from the released reward matrix; common random numbers pair policies within each instance. The original 30-user, two-subset study is nested in this expansion, not an independent replication.

Two ratios answer different questions. We report $\overline{R}_{\text{policy}}/\overline{R}_{\text{baseline}}$ for average loss, and preserve the original tables' $\overline{R_{\text{policy}}/R_{\text{baseline}}}$ as a separate measure of relative instance performance. Small baseline regrets can dominate the latter. For uncertainty and tail summaries, the three subsets are first averaged within each user, then 120 users are bootstrapped with 2,000 paired draws. These intervals condition on the common tuning population and the estimated reward matrix.

Oracle-certified I-mf pooling has mean regret 0.640 times the matched fixed-confidence KL-UCB baseline, but 1.456 times Bernoulli TS. Its mean paired instance ratio to fixed KL-UCB is 0.802. These are compatible conclusions: certification helps within the conservative confidence-based family, while it does not establish superiority over a strong sampling baseline. Energy and global 90th-percentile widths remain wide; on I-mf their fractions of rejectable suboptimal components are 0.209 and 0.146, against 0.594 for oracle widths. The global 50th-percentile width is useful on average but covers only 47.3% of I-mf components. A 90% cross-user quantile is neither a 95% simultaneous guarantee nor an anytime certificate for a new user. Per-user diagnostic coverage is evaluated at the true user level, whereas the policy selects widths after a noisy, paid warm-up; the diagnostic is not realized warm-up coverage.

Topology versus shrinkage. Real I-mf and its degree-preserving rewiring have almost identical spectral-TS mean regret: relative difference +0.29%, paired user interval $[-4.13\%, 4.64\%]$. An interval crossing zero is inconclusive, not evidence of equivalence. Oracle-certified pooling gains 8.45% over the rewired graph, interval $[7.38\%, 9.46\%]$ improvement. Graph-free shrinkage TS has

Policy	Mean / TS	95% CI	Mean / fixed KL	Paired / fixed KL	Valid widths
Graph-free shrinkage TS	0.688	[0.644, 0.734]	0.302	0.902	–
Spectral TS, real I-mf	0.794	[0.723, 0.889]	0.349	1.168	–
Spectral TS, rewired I-mf	0.792	[0.722, 0.880]	0.348	1.164	–
SP-KLUCB, oracle	1.456	[1.393, 1.518]	0.640	0.802	100.0%
SP-KLUCB, energy	2.184	[2.093, 2.274]	0.960	0.977	100.0%
SP-KLUCB, global 90%	2.244	[2.146, 2.334]	0.986	0.992	89.7%
SP-KLUCB, global 50%	1.580	[1.521, 1.634]	0.695	0.825	47.3%
SP-KLUCB, user 90%	2.153	[2.061, 2.237]	0.947	0.969	89.3%
SP-KLUCB, user 75%	1.864	[1.775, 1.945]	0.819	0.898	73.6%
SP-KLUCB, zero width	0.842	[0.791, 0.899]	0.370	0.622	0.0%

Table 11: Completed Setting A v2: 120 held-out users, three subsets each, 300 videos, $T = 20{,}000$. Ratios of mean losses differ from means of instance ratios. Intervals bootstrap users after averaging subsets. User-calibration validity is evaluated at the true user level, not the realized noisy warm-up; it is diagnostic, not a simultaneous certificate guarantee.

13.39% lower mean regret than real I-mf spectral TS, interval $[4.50\%, 22.10\%]$ improvement. Specific topology therefore helps the oracle pooling geometry in this setting, while much of the uncensored smoothing gain is explained by shrinkage. Strong I-coeng spectral TS has a 90th-percentile user-averaged regret ratio of 4.70 to TS; 26.7% of users exceed twice TS. Tail risk persists, but the older 18.78 instance-tail number belongs to a different cohort/subset design and is not the expansion's headline.

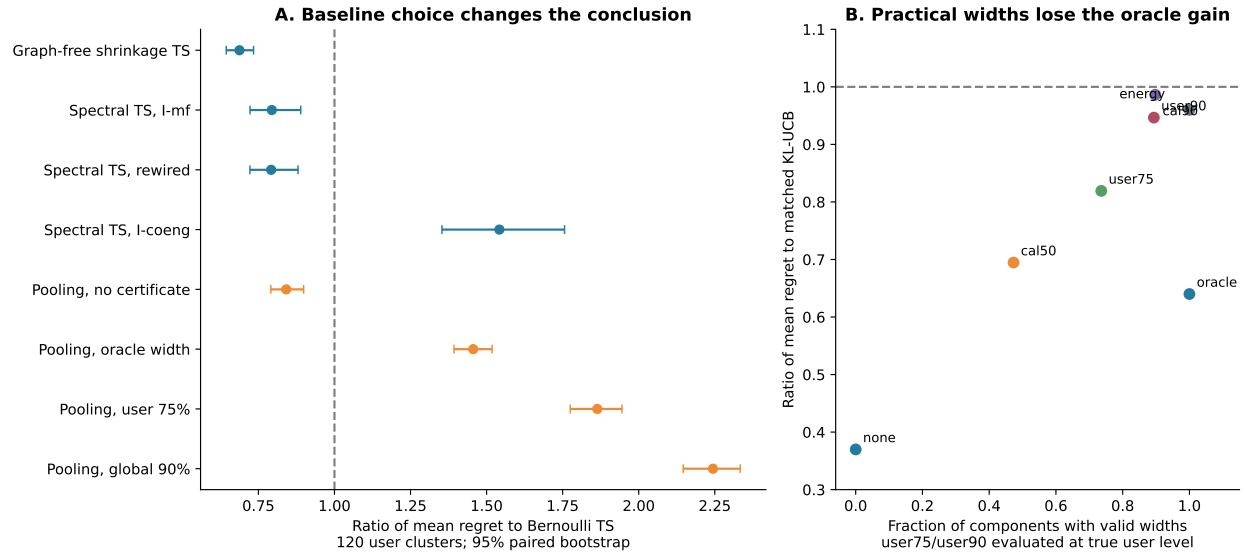


Figure 7: Completed static expansion. Left: ratios of mean regret to Bernoulli TS with user-cluster intervals. Right: empirical component validity against performance relative to the matched conservative KL-UCB baseline. True-user-level calibration diagnostics do not certify the noisy warm-up rule.

7.2 Chronological KuaiRec: persistence survives, topology is unresolved

The corrected daily study rebuilds co-engagement and IDF using only non-evaluation-user interactions before each test cutoff. Eligibility requires at least ten plays on every fitting day and does not condition on future survival. Five panels of 49–50 videos enter each of three disjoint 11-day windows, days 31–41, 42–52 and 53–63. Hyperparameters are selected on days 20–30 and then frozen. Five policy seeds are averaged within each of the 15 panel-window units. The source development panels are drawn using origin-30 eligibility and graph communities; development selection is therefore conditional on that cohort, although the held-out test cutoffs exclude future interactions. The source's policy RNG also uses Python's process-dependent hash, which should be replaced by a stable identifier for exact rerun reproducibility.

Availability is unknown before selection. A chosen video with no plays provides no observation. For retrospective scoring its slot receives the day's median available reward; the oracle selects the best available slate. This convention is part of the monitoring proxy, not an observed platform reward or an online treatment effect. Seventeen test cells are unavailable; 201 unavailable selections occur across 1,950 policy trajectories. Intervals resample the three windows and then their panels. With only three windows and shared global shocks/fits, they are conditional sensitivity summaries rather than broad platform confidence guarantees.

Policy	Loss, k=1	/ IID TS	Loss, k=5	/ IID TS
IID TS	0.4703	1.000	0.3280	1.000
Static	0.4756	1.011	0.3484	1.062
Last value	0.2309	0.491	0.1411	0.430
EWMA	0.2283	0.485	0.2391	0.729
Temporal greedy	0.2334	0.496	0.1495	0.456
Joint greedy	0.2209	0.470	0.1444	0.440
Temporal UCB	0.2399	0.510	0.1696	0.517
Joint UCB	0.2383	0.507	0.1504	0.459
Joint UCB, rewired	0.2351	0.500	0.1525	0.465
Joint UCB, common factor	0.2362	0.502	0.1559	0.475

Table 12: Chronological KuaiRec: observed-oracle logit completion-rate loss per day per selected slot. Five panels in each of three disjoint 11-day windows; policy seeds are averaged within each panel-window. Unavailable selections receive the specified retrospective median fallback for scoring and no feedback.

Joint UCB has 49.3% less loss than iid TS at one selection/day and 54.1% less at five, with relative-difference intervals $[-84.8\%, -13.4\%]$ and $[-67.0\%, -42.6\%]$. Against last observed value, however, its differences are $+3.2\%$ and $+6.6\%$, with intervals crossing zero. Joint greedy versus last value is likewise inconclusive. This supports persistence-aware decisions, but does not establish that a full fitted model outperforms the simplest persistence rule. At five selections/day, joint UCB improves on temporal-only UCB by 11.3%, interval $[5.3\%, 20.4\%]$ improvement. The real graph versus rewiring (-1.4%) and versus a common factor (-3.5%) are inconclusive. Correlated-state inference helps some comparisons; the actual co-engagement topology has no established incremental decision benefit here.

7.3 Historical Wikipedia: the full-year extension is panel-dependent

The historical astronomy and football panels each contain 24 fixed articles and pre-2024 hyperlink matrices. All policies receive the same complete 2024 initialization, 8,784 readings per panel, and

selectively observe the 365-day 2025 field without a full-field test refresh. Policy hyperparameters are fixed using 2024 development data. The extension overlaps the earlier 181-day study and therefore cannot be counted as another independent Web history. Five stochastic seeds quantify algorithmic randomization conditional on each observed field; date-block sensitivities at 7, 14 and 28 days address temporal dependence within those fields.

Policy	Astronomy	Football
IID TS	0.0539	0.2203
Static	0.0535	0.2215
EWMA	0.0414	0.2321
Temporal greedy	0.0544	0.1302
Joint UCB	0.0404	0.1455

Table 13: Full-year Wikipedia extension: mean observed-oracle $\log(1+\text{views})$ loss per selected slot, $k = 5$, 365 test days in 2025. Each 24-article panel receives the same 8,784 historical readings in 2024. These are the same historical panels as the earlier half-year study, not independent platform replications.

At five readings/day, astronomy joint UCB is 25.1% below iid TS, but EWMA is only 2.6% above joint UCB and their comparison is inconclusive. In football, temporal greedy is 40.9% below iid TS and has lower loss than joint UCB. At one reading/day, static ranking and iid rules remain strong. These results replace any statement that full temporal modelling or graph filtering wins on every panel or budget. The older 150-article one-domain and six-community Wikipedia panels used retrospective popularity/completeness selection and current hyperlink snapshots; their shock-correlation and policy results remain diagnostics, not chronological replications.

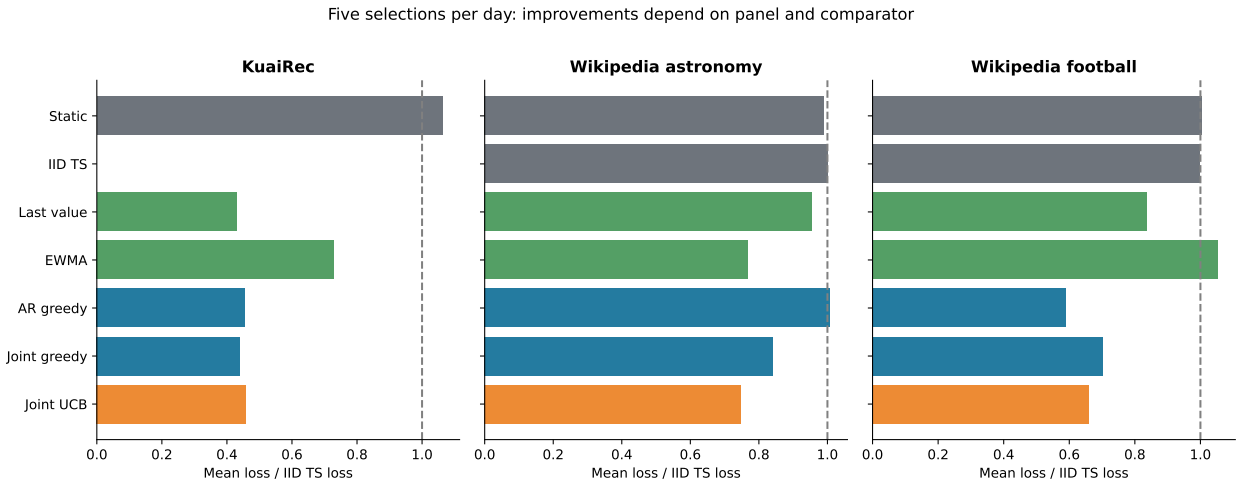


Figure 8: Five readings per day, normalized within each panel by iid-TS loss. KuaiRec uses logit completion-rate loss; Wikipedia uses log-view loss. Ratios aid within-panel comparisons and are not a common cross-platform reward scale.

7.4 Bridge validity, misspecification and the cost of useful decisions

Fresh general-AR confirmation. After a 480-run screen over AR(1), AR(7), AR(20) and heterogeneous orders, a selected AR(20) setting was confirmed on 100 fresh paired worlds per

learner, with an independent fitting prefix. Known-model coverage is compatible with the nominal 5% target; plug-in estimation and covariance inflation remain empirical heuristics. Wrong lag order and diagonal innovations are combined in this confirmation and cannot be attributed separately.

Learner	Anytime failures	95% Wilson interval	Final maximum width
Known model	1/100	[0.002, 0.054]	5.01
Fitted AR(20)	5/100	[0.022, 0.112]	4.92
Fitted, Q times 1.5	3/100	[0.010, 0.085]	5.88
AR(1), diagonal shocks	40/100	[0.309, 0.498]	2.72

Table 14: Fresh AR(20) confirmation after screening: six arms, 800 rounds, burst length 25, an independent 365-round fitting prefix, 100 paired worlds. The last row bundles wrong lag order and diagonalized innovations. Covariance inflation is an empirical heuristic.

Other robustness runs distinguish missed weekly lags and heavy tails: AR(7) truth fitted as AR(1) gives 20–73% violations across policies, while variance-normalized Student-t_3 shocks reach 17%. Short, 100-round fitting prefixes under-cover at high persistence; increasing fitted persistence by two estimated standard errors helps only partially. These stress tests do not refute a theorem assuming correct Gaussian dynamics, and they do not validate learned-model confidence merely because some fitted configurations cover.

Wikipedia-calibrated worlds. The real-dynamics certificate study contains 4,800 runs: 30 articles, six communities, fitted AR(7), horizon 730 days, slates of five or ten, known or fitted dynamics, SP-UCB or successive elimination, and 100 worlds per cell. Gaussian innovations provide the centered reference; independent empirical-residual resampling and weekly block resampling are additional diagnostics. The block stress test violates temporal innovation independence. All pooling widths are oracle widths of the declared mean vector, not learned practical certificates.

SP-UCB confidence	Violations, k=5	Mean regret, k=5	Violations, k=10	Mean regret, k=10
Known AR(7)	0/100	441.0	0/100	466.6
Fitted AR(7)	0/100	459.4	0/100	506.4
Fitted AR(1)	0/100	210.6	0/100	511.1
IID confidence	56/100	37.8	94/100	44.6

Table 15: Centered Gaussian worlds calibrated to 30 Wikipedia articles: 730 days, 100 worlds per cell, oracle community widths. Violations check arm/component upper bounds; mean regret sums slate gaps over the horizon. Low empirical violation rates do not establish a fitted-model theorem.

Gaussian innovation bounds record zero upper-bound violations in each displayed 100-world cell, but zero of 100 has a Wilson upper failure bound of 3.7%, not zero population risk. The iid rule records 56/100 violations for five-slot SP-UCB and 94/100 for ten slots, while its mean regret is much lower. Innovation-based elimination retains all 30 arms; iid elimination has 78–99% upper-bound violations but does not remove the best article in these large-gap Gaussian worlds. The diagnostic does not audit lower-bound coverage or preservation of every top-k arm. Thus invalid coverage need not imply a wrong decision in every environment, and empirical validity does not imply useful rejection or lower finite-horizon regret.

At day 730, a separate five-world random-slate diagnostic reports median arm-interval widths 1.739 for the nuisance plug-in construction, 0.703 for the joint ellipsoid and 0.070 for uncertified

pointwise ±2 posterior standard deviations. The best–second gap is 0.135 and the ellipsoid coefficient is about 20. This identifies confidence construction as a practical bottleneck; pointwise posterior precision is not itself an anytime guarantee.

Residual-centering audit. Empirical residual vectors were not recentered before resampling. Their largest mean coordinate is 0.001534; under the fitted AR polynomial this shifts long-run levels by up to 0.01936 on the rescaled axis. The best article and top-five/top-ten sets remain unchanged, but one of six supplied oracle component widths is too small for the shifted levels. Consequently those resampled cells are implementation diagnostics, not clean nominal-coverage confirmation for the stated fixed means. The Gaussian cells do not have this mean shift. Future resampled replications should center residuals and regenerate certificates and mean-regret targets consistently.

Irreversible failure and finite-horizon regret. In the controlled bursty-elimination follow-up, iid confidence removes the best arm in 28–89% of selected high-persistence configurations, leaving 0.14–0.26 loss per subsequent round; innovation confidence records no such removals but often eliminates nothing through 20,000 rounds. A separate two-arm confirmation finds iid failure 30/100 and 47/100 at persistence 0.97 and 0.995 versus zero innovation failures. At the measured horizons the invalid rule can still have lower average regret. These are finite-horizon safety/usefulness tradeoffs; the observed curves alone are not an asymptotic linear-regret proof or evidence that the valid policy eventually wins.

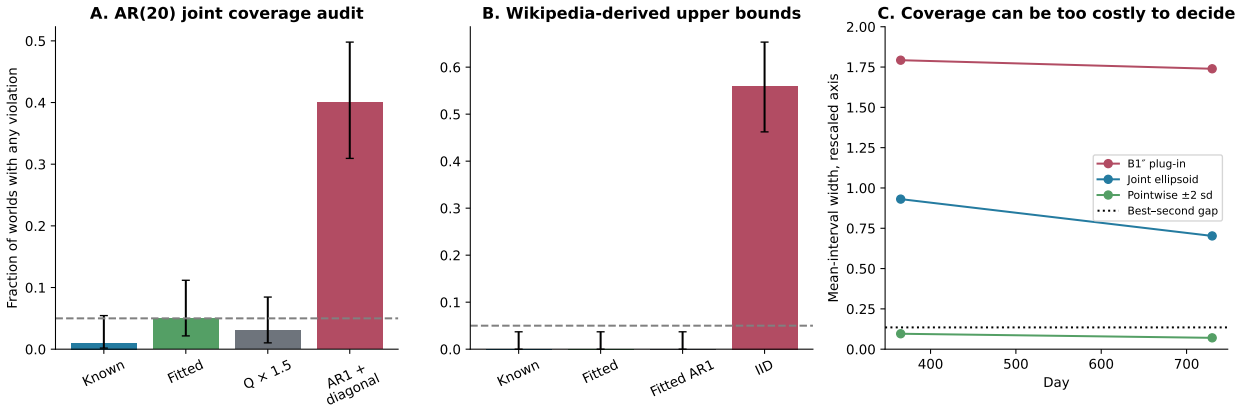


Figure 9: Coverage and usefulness require different evidence. Left: general-AR joint-coverage confirmation. Middle: upper-bound violations in centered Wikipedia-calibrated Gaussian worlds. Right: widths of distinct confidence constructions; the pointwise curve has no anytime coverage claim.

7.5 Long-run information and spatial contrast checks

Full autocovariances matter. On the existing 150-article Wikipedia panels, seven-day block-mean variance is 3.6–5.3 times an iid calculation. AR(7)/AR(20) prefix fits predict substantially more of this inflation than a lag-one AR(1) shortcut, which underpredicts long-block inflation by factors 1.2–3.3 in the reported diagnostics. Those cohorts are retrospective; the fits underpredict for roughly two-thirds of articles, and this diagnostic does not calibrate a real-data confidence theorem.

Pair	Simultaneous variance ratio	Staggered variance ratio
Equal variances, phi=0.9	0.100	0.106
Unequal standard deviations 1, 2	0.280	0.284
Persistence 0.9 and 0.5	0.383	0.406
Equal, iid	0.109	1.000

Table 16: Contrast variance relative to zero innovation correlation, 365 observations per arm, innovation correlation 0.9. The ideal equal-pair, zero-sensor-noise factor is 0.1; these calculations include independent sensor noise of standard deviation 0.1. Heterogeneous pairs and staggered schedules require their own conditional covariance. The staggered design uses a different calendar length; its ratios do not establish equal-budget policy dominance.

The equal-arm $(1-\rho)$ contrast factor is a restricted zero-sensor-noise calculation. The tabulated finite-sample cases include independent sensor noise of standard deviation 0.1, explaining the iid simultaneous ratio 0.109 instead of 0.100. With heterogeneous persistence or marginal variance, use the long-run covariance Ω and contrast variance $\Omega_{ii} + \Omega_{jj} - 2\Omega_{ij}$, adding sensor variance and finite-calendar corrections when needed. At correlation 0.9, unequal pairs retain factors around 0.28 and 0.38 rather than 0.10. Staggered persistent observations retain substantial covariance benefit in the tested schedules; staggered iid observations retain none. This is schedule-specific evidence, not a theorem that simultaneity is unnecessary in every partial-observation design. Independent Monte Carlo checks over eight equal-pair cells and 20,000 worlds per cell agree with the exact finite-sample variances within 1.8%.

An exploratory 16-arm factorial holds stationary marginal variance fixed and changes only the learner's graph in corruption controls. At persistence 0.9, correlation strength 0.8 and small mean spread, joint UCB minus matched temporal UCB is -0.039 latent-oracle loss per round, paired interval $[-0.070, -0.008]$ over 20 worlds. The larger-spread contrast is inconclusive. Corrupting the learner graph worsens the selected joint-UCB contrasts while holding the true world fixed. This supports a spatial mechanism in a controlled regime; the planned 32–64-arm general-AR, 100-world confirmation remains unfinished.

7.6 Open Bandit Dataset: a mean-channel application with sparse clicks

Feasibility and protocol. The three full uniform-random logs contain 2,691,861 impressions, with 80, 34 and 46 items in the All, Men and Women campaigns. Click rates vary across items; daily dispersion is about twice a binomial working-model expectation, but residual lag-one correlations are -0.14 to -0.22 by day and approximately zero in six-hour bins. These measurements do not establish a persistent fluctuation channel at the available resolution. Attribute groups come from released retrospective metadata; audience-response groups use days 1–3 only.

The primary application is explicitly semi-synthetic: day 4–7 point estimates define fixed click means and the learner receives simulated binomial feedback. Graph-free empirical-Bayes TS has the lowest observed mean regret among the tested rules, about 0.206 and 0.290 of conservative fixed-level KL-UCB. Oracle pooling yields smaller, graph-dependent gains; standard log-t KL-UCB and ordinary Beta TS are necessary strong controls. Uncertified audience pooling in All never selects the true-best item in the final 10% of these simulations. This illustrates dilution risk, but is not a real deployment outcome. Prefix ranges cover held-out component ranges in only 25–70% of the original component instances and cannot be called valid certificates.

Policy	All: ratio to fixed KL-UCB	Women: ratio to fixed KL-UCB
Standard log-t KL-UCB	0.684	0.746
Beta TS	0.353	0.382
Graph-free empirical-Bayes TS	0.206	0.290
Oracle pooling, attributes	0.837	0.982
Oracle pooling, audience	0.702	0.965
Prefix pooling, attributes	0.862	1.011
Prefix pooling, audience	0.775	0.809

Table 17: OBD semi-synthetic study: day 4–7 estimated click rates treated as fixed means; components and priors from days 1–3; two million simulated impressions in batches of 100, 20 seeds. Attribute metadata have retrospective provenance. The conservative fixed-confidence KL-UCB comparator is distinct from standard log-t KL-UCB.

Truth uncertainty and drift sensitivity. A separate audit crosses two campaigns, pooled positions or position 1, and prefixes of two, three or four days, producing 12 splits and 72 graph/width comparisons. With a three-day prefix and pooled positions, the empirically best item's posterior probability of being truly best is 0.627 in All and 0.553 in Women under independent Jeffreys rate posteriors. Thus two million simulated impressions do not make the estimated real-data truth precise. Simultaneous binomial prefix intervals enlarge component widths; for the audience graph, posterior probability that all held-out ranges fit rises from zero to 0.915 in All and 0.686 in Women. These are conditional rate-posterior sensitivities under a binomial working model, not frequentist guarantees against future drift. The audit runs no additional bandit trajectories.

Actual censored-log pilot. A separate executed pilot contains 142 policy trajectories over the final 72 hours of the Men's position-1 logs. Each learner chooses before the hour, sees only matching randomized clicks/impressions and advances calendar time through empty bins. The first three dates fit the learner, the fourth is development and the last three are test. About 91.7% of training hourly item bins contain zero clicks, and test trajectories obtain roughly 10–16 clicks. Inverse-propensity summaries are therefore underpowered for a convincing ranking. This pilot is actual logged-feedback evidence for a sparse censored learner; it neither reproduces a deployed full-feedback learner nor validates a Gaussian click filter, permanent-mean PCS or a true latent-state oracle regret.

7.7 What the updated evidence supports

The strongest agreement is with conditional mechanisms: useful pooling depends on supplied widths relative to gaps; persistence changes mean information and can invalidate iid confidence under bursty exposure; correlated shocks can reduce particular contrast variances; and prediction, mean learning and final identification require different allocation objectives. No experiment converts a known-model theorem into a fitted-model guarantee. The tested practical certificates are either wide, empirically incomplete, or both. Persistence-aware decisions improve some real-panel comparisons against iid rules, while last-value, EWMA, static ranking and graph-free shrinkage remain essential comparators. We do not claim universal superiority of the proposed algorithms, an established topology advantage on the dynamic panels, or an online platform deployment.

The next method priority is confidence that remains useful after dynamics estimation and simultaneous coverage are charged. The next empirical priority is a broader, predefined chronological Wikipedia panel set and the focused general-AR factorial confirmation. These gaps should gov-

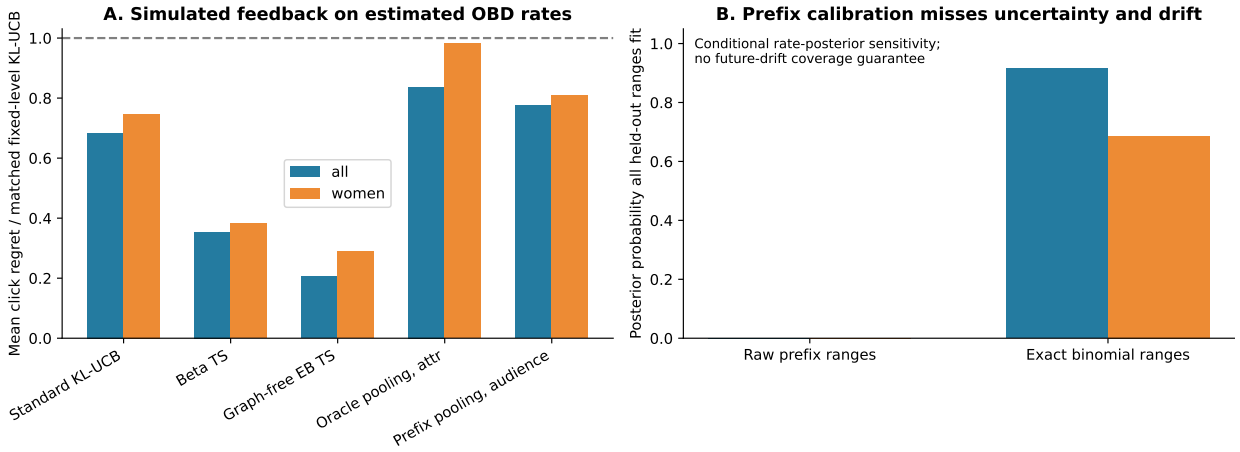


Figure 10: OBD evidence. Left: simulated-feedback regret against a conservative matched KL-UCB baseline, retaining standard KL-UCB and Beta TS. Right: conditional held-out rate-posterior range sensitivity, which does not provide a future-drift guarantee.

ern claims in the condensed paper rather than adding more seeds to already well-covered narrow settings.

Source provenance. All new tables are generated from saved results by the accompanying `build_latest.py`; `source_manifest.json` fingerprints their inputs. The quantitative review is under `experiments/report_2026_10_11/`, and the narrative experiment report under `experiments/report/`. Repeated analyses and copied result snapshots are not independent experiments. Frozen technical appendices retain their source history; the qualifications in the present main text govern the updated empirical interpretation.

8 Lessons, limitations and open problems

A checklist for practitioners.

1. *Measure before trusting.* Compare each candidate graph with its rewiring on held-out data: neighbour agreement after removing popularity (mean channel) and held-out likelihood of shocks (fluctuation channel).
2. *Audit pooling widths and strong controls.* Wrong widths can cause dilution failures; wide valid widths remove much of the benefit. Compare oracle and practical widths separately, and retain Bernoulli TS and graph-free shrinkage.
3. *Account for persistence in every confidence statement,* especially when exposure is bursty. Use innovation-regression certificates for any irrevocable decision.
4. *Match the policy to the objective.* Current reward, permanent-mean regret and correct final selection need different information. Test modelling against last value, EWMA and discounting before attributing a gain to the full filter.
5. *Check what decides the ranking.* If long-run levels dominate fluctuations, a graph in the shock model will barely change decisions even when it fits shocks better.

Limitations. Static KuaiRec and OBD treat noisy data-derived rates as simulation truth. The daily panels emulate exogenous monitoring, not promotion lift. Legacy cohort/link look-ahead, the corrected KuaiRec fallback metric and its three-window uncertainty, fitted dynamics, residual-centering errors in empirical-resampling worlds and underpowered logged clicks are disclosed in Section 7. Correct-model guarantees do not transfer automatically to fitted likelihoods or quantile widths. B1′ assumes independent shocks; correlated-noise regret needs realized-information growth rather than an unverified per-pull factor.

Open problems. (1) Practical component widths and contrast confidence that are both calibrated and useful; level-matched quantiles do not close this gap. (2) Information-growth constants under partial correlated sampling and geometry-aware elimination under persistence. (3) Confidence under estimated dynamics and drift. (4) Broader predefined chronological panels and the focused general-AR factorial confirmation. (5) Evaluation with sufficient real logged signal and an explicitly specified platform utility.

A Proofs of the bridging results

B1 (Theorem 3). Define $U_i = \hat{\mu}_i + \beta_t \|e_i\|_{J_t^{-1}}$ and $U_c = \min\{\max_{i\in C_c} U_i, \min_{w\in\Delta(C_c)}(w^\top\hat{\mu} + \beta_t\|w\|_{J_t^{-1}}) + \varepsilon_c\}$. *Validity:* Theorem 2 holds for all contrasts simultaneously, hence for every e_i and every w in every simplex; for $w \in \Delta(C_c)$, $w^\top\mu \geq v_c - \varepsilon_c$, so both terms of U_c bound $v_c = \max_{i\in C_c}\mu_i$. Minimizing over w is free because the event does not depend on w. *Regret:* if c is selected at round t, then $\mu^\star \leq U_c \leq v_c + \varepsilon_c + 2\beta_t\|w\|_{J_t^{-1}}$, so $\min_w \|w\|^2_{J_t^{-1}} \geq (\Gamma_c - \varepsilon_c)^2/(4\beta_t^2)$; the information-growth condition makes this at most $1/(\iota_c(N_c-1))$. Arms are identical, and summation follows Theorem 1. *AR(1) constant:* with independent arms the information is diagonal. The coefficient of μ_i in a pull of arm i is $1-\varphi^h s$, where $h \geq 1$ is the gap since the arm's last observation and $s \in [0,1]$ is the filter's estimate after unit residuals (each filter step is a convex combination of φ times the previous estimate and the new residual), so it is at least $1-\varphi$; the predictive variance is at most $V + r$.

B1′ (Proposition 4). With Q_G diagonal and $H = \alpha I$, each u_t has one nonzero entry, so each arm's score $W_i = \sum_{t:a_t=i} u_t\epsilon_t$ is a scalar martingale with quadratic variation $\mathcal{I}_{ii}$. The one-

dimensional method of mixtures gives $|W_i| \le \sqrt{2J_{ii}\log(\sqrt{J_{ii}/\alpha}/\delta')}$ at all times with probability at least $1-\delta'$; the regularization bias is at most $\sqrt{\alpha}$. The pooled statistic q_c/J_c with $J_c = \alpha + \sum_{i\in C_c} \mathcal{I}_{ii}$ estimates the information-weighted average of the pulled arms' means, which is at least $v_c - \varepsilon_c$, with a scalar martingale error. A union bound over $N+M$ statistics with $\delta' = \delta/(N+M)$ completes the proof.

B1″ (Proposition 5). Let $G = J - \alpha I$. Then $q_i - \sum_{j\neq i} G_{ij} m_j = G_{ii}\mu_i + \sum_{j\neq i} G_{ij}(\mu_j - m_j) + W_i$, where $W_i = \sum_t u_{t,i}\epsilon_t$ is a scalar martingale with quadratic variation G_{ii}. On the event E that all scalar mixture bounds hold, and given intervals $[l_j, h_j] \ni \mu_j$ with midpoints m_j and half-widths ρ_j, the error of $\hat{\mu}_i = (q_i - \sum_{j\neq i} G_{ij} m_j)/J_{ii}$ is at most $(\sqrt{2J_{ii}\log(\sqrt{J_{ii}/\alpha}/\delta')} + \alpha + \sum_{j\neq i} |G_{ij}|\rho_j)/J_{ii}$. Starting from $[0,1]$, which contains the truth, induction keeps every interval true-containing on E. For components, write $\mu_i = \theta_c + d_i$ with θ_c the midrange and $|d_i| \le \varepsilon_c/2$; the design $P^\top u_t$ (P the membership matrix) gives the same argument for θ, with a deterministic deviation term $\sum_i |(P^\top G)_{ci}|\varepsilon_{c(i)}/2$, so $\hat{\theta}_c + \rho_c + \varepsilon_c/2 \ge v_c$. With independent shocks G is diagonal, the cross terms vanish and the deviation term equals $\varepsilon_c/2$ times the information, recovering SP-UCB's structure. In implementation, upper bounds must not be clipped at 1, since ties created by clipping are broken by index.

B2 (Proposition 6). Within one block of b consecutive pulls the autocovariances are $V\varphi^{|s-u|}$; summing gives $\mathrm{Var}(\sum) = V[b\frac{1+\varphi}{1-\varphi} - \frac{2\varphi(1-\varphi^b)}{(1-\varphi)^2}]$. Divide by b^2 and use Gaussian tails for the exact stated miscoverage. For arbitrary fixed observation times $\tau_1, \ldots, \tau_n$, the variance is $Vn^{-2}\sum_{s,u} \varphi^{|\tau_s - \tau_u|}$. Dropping nonnegligible cross-block terms is not exact. Run-level adaptive simultaneous violations are measured separately from this fixed-block calculation.

B3 (Proposition 7). After the initial lags, the full-field innovation transform $X_t - \sum_k \mathrm{diag}(a_k) X_{t-k} = D_c\mu + \eta_t$ adds information $D_c Q_G^{-1} D_c$ per field [S]. With common coefficients this identifies the long-run covariance Q_G/c^2. A subset observation marginalizes unobserved coordinates, so its covariance is Q_{SS} and its precision $(Q_{SS})^{-1}$; the block $(Q_G^{-1})_{SS}$ instead conditions on unavailable coordinates. Equal-pair contrast variance gives the restricted factor $1-\rho$; heterogeneous pairs use the full long-run covariance. Inverse pooled variance increases through the sum of arm information: one scalar reading has the independent-arm information lower bound, not a component-size multiple of it. General partial schedules need the conditional filter and a separately justified information-growth condition.

B Data and graph construction

KuaiRec. Small matrix: 1,411 users × 3,327 videos, 4,676,570 interactions, 17,827 blocked pairs. Big matrix: 7,176 × 10,728, 12,530,806 interactions; no overlap with the small matrix. Social network: 472 users, 146 in the evaluation set, 47 edges among them. Reward R1 = min(watch ratio, 5)/5 (median watch ratio 0.77; 4.6% above 2); R3 counts a watch ratio above 2 as a like. Graphs are union-kNN; Laplacians are scaled to mean degree one; nulls are degree-preserving rewirings and Erdős–Rényi graphs. U-coeng: Jaccard on fully watched non-evaluation videos. U-coeng-idf and U-coauthor: cosine on IDF-weighted engagement by video or by creator. U-mf and I-mf: PureSVD (rank 64) on leakage-free splits. U-cotag: per-tag taste. U-cotime: hour-of-day profile. U-feat, U-geo (same city 1, province 0.3), U-demo: profile fields. I-coeng:

IDF-weighted engagement from non-evaluation users. I-coauthor: same creator. I-tag, I-cat: tags and three-level categories.

Wikipedia. Public Wikimedia REST API (daily user page views, all access, 2022–2025) and MediaWiki links API. One domain: the 150 most-viewed articles linked from "List of programming languages" with complete data (512 of 691), 3,037 hyperlinks. Six domains: 25 articles each from NFL teams, NBA teams, MLB teams, Premier League clubs, UN member states and programming languages, 1,990 hyperlinks (84% within a domain).

C Reproducibility

Each experiment has its own directory under `experiments/` (`kuairec`, `wikipedia`, `simulations`) holding its design, code, results and report. From the repository root:

```
# Q1, mean channel (KuaiRec)
python -I experiments/kuairec/code/{data,graphs,alignment}.py
# Q2, certified pooling (KuaiRec)
python -I experiments/kuairec/code/{setting_a,setting_a2,setting_b}.py
# Q3, certificates under persistence
python -I experiments/simulations/certificates_under_persistence/code/coverage.py \
    --tag {_v2,_elim,_scalar,_plugin}
# Q4, benchmark, KuaiRec daily, Wikipedia
python -I experiments/simulations/spatiotemporal_benchmark/code/toy.py [--extra]
python -I experiments/kuairec/code/daily.py [--extra | --sparse]
python -I experiments/wikipedia/code/{collect,collect_multi}.py
python -I experiments/wikipedia/code/attention.py [--dataset multi] [--diagnose]
# theory-to-data analyses and figures
python -I experiments/paper/{deepdive,paper_figures_v2}.py
```